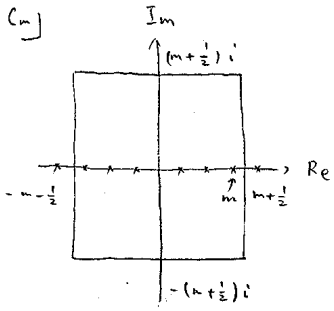
複素関数論

1. $f(z) = \dfrac{1}{z^2+a^2}$, $g(z) = \dfrac{\pi}{\tan \pi z}$



$$\int_{C_m} f(z)\, g(z)\, dz = 2\pi i \sum_{n=-m}^{m} \frac{1}{n^2+a^2} + 2\pi i \left[\frac{g(ia)}{2ia} + \frac{g(-ia)}{-2ia} \right]$$

$$= 2\pi i \left[\frac{1}{a^2} + 2\sum_{n=1}^{m} \frac{1}{n^2+a^2} - \frac{\pi}{a} \coth(\pi a) \right] \xrightarrow{(m\to\infty)} 0$$

$$\tan(ix) = i \tanh(x)$$

以下 $|\int| \to 0$ を示す

$$\left|\frac{1}{z^2+a^2}\right| \cdot \left|\frac{\pi}{\tan \pi z}\right| \qquad \int |dz| \sim m$$

$$\leq \frac{1}{m^2} \qquad < \pi$$

$z = x+iy$

$x = m+\frac{1}{2}$; $\tan(\pi z) = \tan\left(m\pi + \frac{\pi}{2} + i\pi y\right) = \tan\left(\frac{\pi}{2} + i\pi y\right) = -\dfrac{1}{\tan i\pi y} = -\dfrac{1}{\tanh(\pi y)}$

$$\therefore \left|\frac{\pi}{\tan(\pi z)}\right| = |\pi \tanh(\pi y)| \leq \pi$$

$y = m+\frac{1}{2}$; $\tan(\pi z) = \tan\left(\pi x + i\left(m\pi + \frac{\pi}{2}\right)\right) \to -\frac{1}{i}$ $(m \to \infty)$

以上より

$$\sum_{n=1}^{\infty} \frac{1}{n^2+a^2} = \frac{1}{2a} \cdot \underbrace{\frac{\pi}{\tanh(\pi a)}} - \frac{1}{2a^2}$$

$$\pi \frac{1 + \frac{(\pi a)^2}{2!} + \cdots}{\pi a + \frac{(\pi a)^3}{3!} + \cdots} = \frac{1}{a}\left(1 + \frac{(\pi a)^2}{2} - \frac{(\pi a)^2}{6} + O(a^4)\right)$$

$$= \frac{1}{2a^2}\left(\frac{1}{3}(\pi a)^2 + O(a^4)\right) \to \frac{\pi^2}{6}$$

∎ 3.の答

2.

$f(z) = \dfrac{1}{(z-a)^2}$　　$z=a$ で 2位の極　　(a は整数ではない)

$|g(z)|$ は C_m 上で有界で $|g(z)| < M$ とする.

$$\left| \int_{C_m} f(z)\, g(z)\, dz \right| \leq \int_{C_m} \frac{M\, dz}{|z-a|^2} \longrightarrow 0 \quad (m \to \infty)$$

$f(z)g(z)$ の $z=a$ での留数は $\lim_{z\to a}\left(\dfrac{d}{dz}\cdot\dfrac{\pi}{\tan \pi z}\right) = -\dfrac{\pi^2}{\sin^2 \pi a}$

$$\therefore\ -\frac{\pi^2}{\sin^2 \pi a} + \sum_{n=-\infty}^{\infty} \frac{1}{(n-a)^2} = 0$$

m次の pole に対して

$$\operatorname*{Re}_{z=a} f(z) = \lim_{z\to a} \frac{1}{(m-1)!}\left(\frac{\partial}{\partial z}\right)^{m-1} (z-a)^m f(z)$$

$$\operatorname*{Re}_{z=a} \frac{1}{(z-a)^2}\cdot\frac{\pi}{\tan(\pi z)} = \lim_{z\to a} \frac{\partial}{\partial z}\cdot\frac{\pi}{\tan(\pi z)} = -\pi \frac{1}{\tan(\pi z)}\cdot\frac{\pi}{\cos^2(\pi z)}\bigg|_{z=a} = \frac{\pi}{\sin^2(\pi a)}$$

Riemann の ζ-関数

$$\zeta(z) = \sum_{n=0}^{\infty} \frac{1}{n^z} \quad (\operatorname{Re} z > 1)$$

$\zeta(1)$ → 発散

$\zeta(2) = \dfrac{\pi^2}{6}$ ← 3の答

4.

$$\int_{C_m} \frac{e^{iz\xi}}{z^2+a^2} \cdot \frac{2\pi i}{e^{2\pi i z}-1}\, dz$$

留数 $z=n$ $\quad \frac{e^{in\xi}}{n^2+a^2}$

$z=ia$ $\quad \frac{e^{-a\xi}}{2ia} \cdot \frac{2\pi i}{e^{-2\pi a}-1} = \frac{\pi}{a} \cdot \frac{e^{-a\xi}}{e^{-2\pi a}-1}$

$z=-ia$ $\quad \frac{e^{a\xi}}{-2ia} \cdot \frac{2\pi i}{e^{2\pi a}-1} = -\frac{\pi}{a} \cdot \frac{e^{a\xi}}{e^{2\pi a}-1}$

$$= \sum_{n=-\infty}^{\infty} \frac{e^{in\xi}}{n^2+a^2} + \frac{\pi}{a^2}\left(\frac{e^{-a\xi}}{e^{-2\pi a}-1} - \frac{e^{a\xi}}{e^{2\pi a}-1} \right)$$

$$= -\frac{\cosh a(\xi-\pi)}{\sinh(\pi a)}$$

$$= \frac{1}{a^2} + 2\sum_{n=1}^{\infty} \frac{\cosh n\xi}{n^2+a^2} - \frac{\cosh a(\xi-\pi)}{\sinh(\pi a)} \to 0$$

$$\therefore \sum_{n=1}^{\infty} \frac{\cosh n\xi}{n^2+a^2} = \frac{\pi}{2a} \cdot \frac{\cosh a(\xi-\pi)}{\sinh(\pi a)} - \frac{1}{2a^2} \quad //$$

☺ $f = \left| \frac{e^{iz\xi}}{e^{2\pi i z}-1} \right| = \left| \frac{e^{i\xi x - \xi y}}{e^{2\pi i x - 2\pi y}-1} \right| = \frac{e^{-\xi y}}{\left| e^{-2\pi y}(\cos 2\pi x + i \sin 2\pi y) - 1 \right|}$

$$= \frac{e^{-\xi y}}{\sqrt{e^{-4\pi y} + 1 - 2e^{-2\pi y}\cos(2\pi x)}}$$

$x = \pm\left(m+\frac{1}{2}\right)$ $\quad f \leqq \frac{e^{-\xi y}}{1+e^{-2\pi y}} \leqq \begin{cases} e^{-\xi y} \leqq 1 & (y \geqq 0) \\ e^{(2\pi-\xi)y} \leqq 1 & (y \leqq 0) \end{cases}$

$y = m+\frac{1}{2}$ $\quad f \leqq \frac{e^{-\xi(m+\frac{1}{2})}}{1-e^{-2\pi(m+\frac{1}{2})}} \longrightarrow \leqq 1 \quad (m \to \infty)$

$y = -\left(m+\frac{1}{2}\right)$ $\quad f \leqq \frac{e^{\xi(m+\frac{1}{2})}}{e^{2\pi(m+\frac{1}{2})}-1} \longrightarrow e^{(\xi-2\pi)(m+\frac{1}{2})} \quad (m \to \infty)$

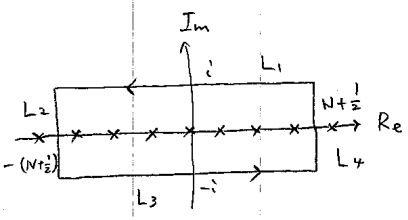
$\leqq 1$

有界

5.

$$\int_{C_N} F(z)\,dz = 2\pi i \sum_{n=-N}^{N} \mathrm{Res}(F, n) = \sum_{n=-N}^{N} e^{-n^2\pi t} \to f(t) \quad (N\to\infty)$$

$$\lim_{z\to n}{}' \frac{z-n}{e^{2\pi i z}-1} e^{-z^2\pi t} = \frac{e^{-n^2\pi t}}{2\pi i}$$



$$|\int_{L_4}| \le 2\left|\frac{e^{-(N+\frac{1}{2}+iy)^2\pi t}}{e^{2\pi i(N+\frac{1}{2})-2\pi y}-1}\right| = \frac{e^{-(N+\frac{1}{2})^2\pi t} e^{y^2\pi t}}{e^{-2\pi y}+1} \to 0 \quad (N\to\infty)$$

L_2 も同様.

L_1 $z = x+i$ で $|e^{2\pi i z}| < 1$ より $F(z) = -e^{-z^2\pi t}\dfrac{1}{1-e^{2\pi i z}} = -e^{-z^2\pi t}\sum_{n=0}^{\infty} e^{2\pi i n z}$

L_3 $z = x-i$ $|e^{-2\pi i z}| < 1$ $F(z) = \dfrac{e^{-z^2\pi t}}{e^{2\pi i z}}\cdot\dfrac{1}{1-e^{-2\pi i z}} = e^{-z^2\pi t}\sum_{n=0}^{\infty} e^{-2\pi i(n+1)z}$

$$= e^{-z^2\pi t}\sum_{n=1}^{\infty} e^{-2\pi i n z}$$

$$\int_{L_1} = -\sum_{n=0}^{\infty}\int_{L_1} e^{-z^2\pi t+2\pi i n z}\,dz = -\sum_{n=0}^{\infty}\int_{\infty}^{-\infty} e^{-\pi t(x-i\frac{n-t}{t})^2-\pi\frac{n^2}{t}}\,dx = \sum_{n=0}^{\infty}\frac{1}{\sqrt{t}} e^{-\frac{\pi n^2}{t}}$$

$$\int_{L_3} = \sum_{n=1}^{\infty}\int_{L_3} e^{-z^2\pi t-2\pi i n z}\,dz = \sum_{n=1}^{\infty}\int_{-\infty}^{\infty} e^{-\pi t(x+i\frac{n-t}{t})^2-\pi\frac{n^2}{t}}\,dx = \sum_{n=1}^{\infty}\frac{1}{\sqrt{t}} e^{-\frac{\pi n^2}{t}}$$

$$\therefore f(t) = \int_{L_1+L_2+L_3+L_4} = \frac{1}{\sqrt{t}}\sum_{n=-\infty}^{\infty} e^{-n^2\pi\frac{1}{t}} = t^{-\frac{1}{2}} f(\tfrac{1}{t})$$

Poisson の和公式

テータ関数 $\vartheta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x} = \frac{1}{\sqrt{x}}\vartheta(\frac{1}{x})$ $\quad x>0$

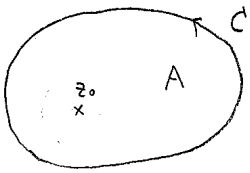
$f(\tau) = e^{-\frac{x}{4\pi}\tau^2}$

$$\int_{-\infty}^{\infty} f(\tau) e^{-in\tau}\,d\tau = \int_{-\infty}^{\infty} e^{-\frac{x}{4\pi}(t+i\frac{2\pi n}{x})^2-\frac{\pi n^2}{x}}\,dt = \frac{2\pi}{\sqrt{x}} e^{-\frac{\pi n^2}{x}}$$

$$\frac{1}{2\pi}\sum_{n=-\infty}^{\infty}\int_{-\infty}^{\infty} f(\tau) e^{-in\tau}\,d\tau = \vartheta(x) = \sum_{n=-\infty}^{\infty} f(2\pi n)$$

6.

$$F(z) := \frac{f(z)-f(z_0)}{g(z)-g(z_0)}\,g'(z) \quad \text{とおく}$$



$F(z)$ は A 内で analytic $(F(z_0)=f'(z_0))$

$$0=\int_C F(z)\,dz=\int_C \frac{f(z)}{g(z)-g(z_0)}\,g'(z)\,dz - f(z_0)\int_C \underbrace{\frac{g'(z)}{g(z)-g(z_0)}}_{\text{pole は } z=z_0 \text{ 留数 } 1 \text{ (1位)}}\,dz$$

$$\therefore\ f(z_0)=\frac{1}{2\pi i}\int_C \frac{f(z)}{g(z)-g(z_0)}\,g'(z)\,dz$$

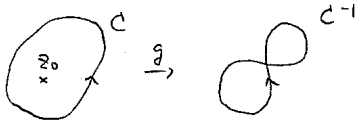
☹ $g(z)-g(z_0)$ の零点は $g'(z)\neq 0$ より z_0 のみ さらに1位である

$t=g(z)$ は零点 $z_0\in C$ を持つ

$$z=z_0+\sum_{n=1}^{\infty}a_n t^n \qquad \int_{C^{-1}}\frac{z}{t^{n+1}}\,dt=2\pi i\,a_n$$

$$\int_{C^{-1}}\frac{z}{t^{n+1}}\,dt=\int_C \frac{z}{g(z)^{n+1}}\,g'(z)\,dz$$

$C^{-1}: t=g(z)$ $z\in C$



$t=0$ は C^{-1} 内にある

☹ $|g(z)|>|g(z_0)|$ (t, 0)

☹ $z_1\neq z_2$ で $g(z_1)=g(z_2)$ となるときは ない

7.

$$y=\frac{x-1}{x^m}=\frac{e^{\ln x}-1}{e^{m\ln x}} \qquad \ln x=0 \to y=0 \text{ より } c_0=0$$

$$a_n=\frac{1}{2\pi i}\oint \frac{\ln x}{y^{n+1}}\frac{dy}{d\ln x}\,d\ln x=\frac{1}{2\pi i}\oint \frac{\ln x}{y^{n+1}}\frac{dy}{dx}\,dx=-\frac{1}{2\pi n i}\oint \ln x\,\frac{d\,y^{-n}}{dx}\,dx$$

$$=\frac{1}{2\pi n i}\oint \frac{dx}{x\,y^n}=\frac{1}{2\pi n i}\oint \underbrace{\frac{x^{mn-1}}{(x-1)^n}}_{x=1 \text{ で } n \text{ 位の pole}}\,dx=\frac{(nm-1)(nm-2)\cdots(nm-n+1)}{n!}$$

留数 $\frac{1}{(n-1)!}\left(\frac{d}{dx}\right)^{n-1}x^{mn-1}\Big|_{x=1}=\frac{\overbrace{(mn-1)(mn-2)\cdots}^{n-1\text{コ}}}{(n-1)!}$

8.

$f(z)$ は $z=a_n$ で1位の pole $a_n\neq 0$

$$\frac{1}{2\pi i}\oint_{C_n}\frac{f(z)}{z-x}\,dz=f(x)+\sum_n \frac{1}{a_n-x}\,\mathrm{Res}(f,a_n)$$

$$=\frac{1}{2\pi i}\oint_{C_n}\frac{f(z)}{z}\,dz+\frac{x}{2\pi i}\oint_{C_n}\frac{f(z)}{z(z-x)}\,dz$$

$$\downarrow \qquad\qquad \downarrow$$

$$f(0)+\sum_n \frac{1}{a_n}\mathrm{Res}(f,a_n) \qquad \frac{|f(z)|}{|z|}\to 0$$

9.

$f(z) = \frac{1}{\sin z} - \frac{1}{z}$ $\quad f(0) = 0$ $\quad z = n\pi$ $(n \neq 0)$ 1位の pole $\quad \mathrm{Res}(f, n\pi) = \frac{z - n\pi}{\sin z - \sin n\pi} \cdot \frac{z - \sin z}{z}\Big|_{z \to n\pi}$

$= \frac{1}{\cos n\pi} = (-1)^n$

10.

$\mathrm{Res}\left(\frac{f'}{f}, a_n\right) = \frac{z - a_n}{f(z) - f(a_n)} f'(z)\Big|_{z \to a_n} = 1$

$\frac{f'(z)}{f(z)} = \frac{f'(0)}{f(0)} + \sum \left(\frac{1}{z - a_n} + \frac{1}{a_n}\right)$

$\ln f(z) - \ln f(0) = \frac{f'(0)}{f(0)} z + \sum \left(\ln(z - a_n) - \ln(-a_n) + \frac{z}{a_n}\right)$

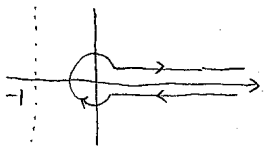
$\ln\left(1 - \frac{z}{a_n}\right) + \frac{z}{a_n}$

$f(z) = f(0)\, e^{\frac{f'(0)}{f(0)} z}\, e^{\sum \left\{\ln\left(1 - \frac{z}{a_n}\right) + \frac{z}{a_n}\right\}}$

$= f(0)\, e^{\frac{f'(0)}{f(0)} z} \prod \left\{\left(1 - \frac{z}{a_n}\right) e^{\frac{z}{a_n}}\right\}$

11. $f(z) = \frac{\sin z}{z}$ $\quad f(0) = 1$ $\quad f'(0) = 0$ $\quad z = n\pi$ $(n \neq 0)$ 1位の零点 $\quad \frac{\sin z}{z} = \prod_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \left(1 - \frac{z}{n\pi}\right) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2\pi^2}\right)$

12.



$\int_D t^{z-1} e^{-t}\, dt =$

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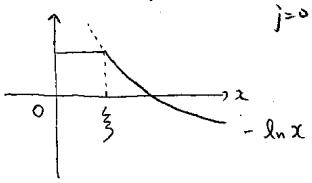
a) $\tilde{g}_{cont}(x) = \begin{cases} g_1(x) & (x<a) \\ g_2(x) + G_0 & (x>a) \end{cases}$ $\qquad G_0 = \lim_{x\to a} g_1(x) - \lim_{x\to a} g_2(x) = g_1(a) - g_2(a)$

$D_x g(x) = D_x \tilde{g}_{cont}(x) - G_0 \delta(x-a) = g'(x) - G_0 \delta(x-a)$ $\qquad g^{(n)}(x) = \begin{cases} g_1^{(n)}(x) & (x<a) \\ g_2^{(n)}(x) & (x>a) \end{cases}$

$D_x^2 g(x) = D_x g'(x) - G_0 D_x \delta(x-a) = g''(x) - G_1 \delta(x-a) - G_0 D_x \delta(x-a)$

$D_x^n g(x) = D_x^{n-1}\left(g'(x) - G_0\delta(x-a)\right) = \cdots = g^{(n)}(x) - \sum_{j=0}^{n-1} G_j D_x^{n-j-1}\delta(x-a)$

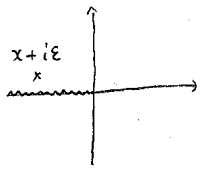
b) $D_x G(x;\xi) = \begin{cases} 0 & (0<x<\xi) \\ -\frac{1}{x} & (\xi<x) \end{cases}$



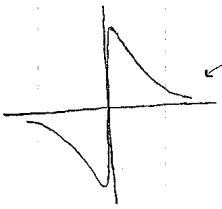
$D_x^2 G(x;\xi) = G''(x;\xi) - \left(0 - \left(-\frac{1}{\xi}\right)\right)\delta(x-\xi) = \left\{ \begin{matrix} 0 & (0<x<\xi) \\ -\frac{1}{x^2} & (\xi<x) \end{matrix} \right\} - \frac{1}{\xi}\delta(x-\xi)$

$L_x G(x;\xi) = (x D_x^2 + D_x) G(x;\xi) = \underbrace{x G'' + G'}_{\parallel\ 0} - \frac{x}{\xi}\delta(x-\xi) = -\delta(x-\xi)$

c) $D_x(\ln x) = \frac{1}{x+i\varepsilon} = P\frac{1}{x} - i\pi\delta(x)$: Dirac の公式



principal value



$= \left(\frac{x}{x^2+\varepsilon^2}\right) - \frac{i\varepsilon}{x^2+\varepsilon^2}$

d) $D_x|x| = \begin{cases} 1 & (x>0) \\ -1 & (x<0) \end{cases}$ $\qquad D_x^2|x| = 2\delta(x)$ $\qquad |x| = x\,\theta(x) - x\,\theta(-x)$

$x\delta(x) = 0 \;\to\; x\delta'(x) = -\delta(x)$

(odd, even)

e) $f(x) = \frac{1}{x\pm i\varepsilon} = P\frac{1}{x} \mp i\pi\delta(x)$

$D_x f(x) = -P\frac{1}{x^2} \mp i\pi\delta'(x)$

$\vdots$

$D_x^n f(x) = (-1)^n n!\, P\frac{1}{x^{n+1}} \mp i\pi\delta^{(n)}(x)$

$\int_{-\infty}^{\infty} dx\, \frac{f^{(n)}(x)}{x+i\varepsilon} = P\int_{-\infty}^{\infty} dx\, \frac{f^{(n)}(x)}{x} - i\pi\int_{-\infty}^{\infty} dx\, f^{(n)}(x)\,\delta(x)$

(注) $\int_{-\infty}^{\infty} f(x)\,\delta'(x-a)\,dx = f(x)\,\delta(x-a)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x)\,\delta(x-a)\,dx = -f'(a)$

$L_x = a(x) D_x^2 + b(x) D_x + c(x) = \underline{D_x(p(x) D_x) + g(x)}$ 標準形

$$\begin{cases} L_x y(x) = f(x) \\ y(0) = y(a) = 0 \end{cases} \Rightarrow y(x) = \int_0^a \underbrace{G(x;\xi)}_{\text{Green 関数}} f(\xi)\, d\xi \qquad \begin{pmatrix} L_x G(x;\xi) = \delta(x-\xi) \\ G(0;\xi) = G(a;\xi) = 0 \quad (0<\xi<a) \end{pmatrix}$$

連続性 $G(x;\xi)_{x=\xi+0} = G(x;\xi)_{x=\xi-0}$

微係数 $\int_{\xi-\varepsilon}^{\xi+\varepsilon} L_x G(x;\xi)\, dx = \int_{\xi-\varepsilon}^{\xi+\varepsilon} \delta(x-\xi)\, dx = 1 \Rightarrow D_x G(x;\xi)\big|_{x=\xi+0} - D_x G(x;\xi)\big|_{x=\xi-0} = \frac{1}{p(x)}$

a) $L_x = D_x^2 \rightarrow u(x) = Ax + B$

$$G(x;\xi) = \begin{cases} \alpha x & (0 \le x < \xi) \\ \beta(x-a) & (\xi < x \le a) \end{cases} \qquad \begin{matrix} \text{連続性} & \alpha\xi = \beta(\xi - a) \\ \text{微分のとび} & \beta - \alpha = 1 \end{matrix} \Bigg\} \begin{matrix} \alpha = \frac{\xi}{a} - 1 \\ \beta = \frac{\xi}{a} \end{matrix}$$

$$G(x;\xi) = \frac{x}{a}(\xi - a)\theta(\xi - x) + \frac{\xi}{a}(x-a)\theta(x-\xi) \Rightarrow G(x;\xi) = G(\xi;x) \text{ 対称}$$

b) $L_x = D_x(x D_x) - \frac{1}{x} \rightarrow u(x) = x^\lambda,\ (\lambda^2 - 1)x^{\lambda-1} = 0,\ \lambda = \pm 1$

$$G(x;\xi) = \begin{cases} \alpha x \\ \beta\left(x - \frac{1}{x}\right) \end{cases} \qquad \begin{matrix} \alpha\xi = \beta\left(\xi - \frac{1}{\xi}\right) \\ \beta\left(1 + \frac{1}{\xi^2}\right) - \alpha = \frac{1}{\xi} \end{matrix} \Bigg\} \begin{matrix} \alpha = \frac{1}{2}\left(\xi - \frac{1}{\xi}\right) \\ \beta = \frac{\xi}{2} \end{matrix}$$

$$G(x;\xi) = \frac{x}{2}\left(\xi - \frac{1}{\xi}\right)\theta(\xi - x) + \frac{\xi}{2}\left(x - \frac{1}{x}\right)\theta(x-\xi)$$

c) $L_x \int G(x;\xi) f(\xi)\, d\xi = \int f(\xi)\, \delta(x-\xi)\, d\xi = f(x)$

$y(x) = \int G(x;\xi) f(\xi)\, d\xi + z(x) \qquad L_x z(x) = 0,\ z(0) = z(1) = 0$

d) $$y(x) = \int_0^a G(x;\xi) f(\xi)\, d\xi = \int_0^x \frac{\xi}{a}(x-a) f(\xi)\, d\xi + \int_x^a \frac{x}{a}(\xi - a) f(\xi)\, d\xi$$

$$= \frac{1}{a}(x-a)\int_0^x \xi f(\xi)\, d\xi + \frac{1}{a} x \int_x^a (\xi - a) f(\xi)\, d\xi$$

e) $$G(x;\xi) = \begin{cases} A\,\mathrm{Ai}(\lambda x)\,\mathrm{Ai}\,\lambda(\xi - a) & (x < \xi) \\ A\,\mathrm{Ai}(\lambda\xi)\,\mathrm{Ai}\,\lambda(x-a) & (x > \xi) \end{cases}$$

微分のとび $\quad A\lambda\,\mathrm{Ai}(-\lambda a) = 1$

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a) $G_\lambda = \sum_n a_n u_n(x) \quad \rightarrow \quad (L_x - \lambda) G_\lambda = \sum_n (\lambda_n - \lambda) a_n u_n(x) = \delta(x-\xi)/\omega(x)$

$$\sum_n (\lambda_n - \lambda) a_n \underbrace{\int_0^a \bar{u}_m u_n \omega \, dx}_{=\delta_{mn}} = \int_0^a \delta(x-\xi) \bar{u}_m \, dx$$

$$(\lambda_m - \lambda) a_m = \bar{u}_m(\xi) \quad \rightarrow \quad a_n(\xi, \lambda) = \frac{\bar{u}_m(\xi)}{\lambda_m - \lambda}$$

b) $$\frac{1}{2\pi i} \int_C G_\lambda(x;\xi) \, d\lambda = \sum_n \frac{1}{2\pi i} \int_C \frac{\bar{u}_n(\xi) u_n(x)}{\lambda_n - \lambda} d\lambda = - \sum_n \bar{u}_n(\xi) u_n(x)$$

$$= - \sum_n \frac{\bar{u}_n(\xi)}{\lambda_n - \lambda} (L_x - \lambda) u_n(x)$$

$$= - (L_x - \lambda) \sum_n \frac{\bar{u}_n(\xi) u_n(x)}{\lambda_m - \lambda} = - \delta(x-\xi)/\omega(x)$$

c) $L_x u_n = D_x^2 u_n(x) = \lambda_n u_n(x)$

$u_n(0) = u_n(a) = 0 \quad \rightarrow \quad \lambda_n < 0 \qquad u_n(x) = C_n \sin\left(\sqrt{-\lambda_n}\, x\right)$

$\sqrt{-\lambda_n}\, a = n\pi \quad \rightarrow \quad \lambda_n = -\left(\frac{n\pi}{a}\right)^2 \qquad (n = 1, 2,$

規格化 $C_n = \sqrt{\frac{2}{a}}$ $\qquad G_\lambda(x;\xi) = \sum_{n=1}^{\infty} \frac{\frac{2}{a} \sin\left(\frac{n\pi}{a} x\right) \sin\left(\frac{n\pi}{a} \xi\right)}{-\left(\frac{n\pi}{a}\right)^2 - \lambda^2}$

$\left(\because \int_0^a \sin^2\left(\frac{n\pi}{a} x\right) dx = \frac{a}{2} \right)$

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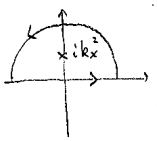
a) $F(D_{x_i})\ G(\vec{r},\vec{r}') = \delta(\vec{r},\vec{r}')$

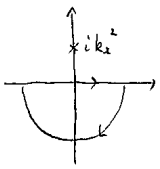
$$F(D_{x_i})\int a(\vec{k})\, e^{i\vec{k}(\vec{r}-\vec{r}')}\, d\vec{k} = \int a(\vec{k})\, F(ik_i)\, e^{i\vec{k}(\vec{r}-\vec{r}')}\, d\vec{k} = \frac{1}{(2\pi)^n}\int e^{i\vec{k}(\vec{r}-\vec{r}')}\, d\vec{k}$$

$$\therefore\ a(\vec{k}) = \frac{1}{(2\pi)^n F(ik_i)}\ ,\quad G_s(\vec{r},\vec{r}') = \frac{1}{(2\pi)^n}\int \frac{e^{i\vec{k}(\vec{r}-\vec{r}')}}{F(ik_i)}\, d\vec{k}$$

b) $F = (ik_x)^2 - ik_y$

$$G_s = \frac{1}{4\pi}\int \frac{e^{i\{k_x(x-x')+k_y(y-y')\}}}{-k_x^2 - ik_y}\, dk_x dk_y = \frac{i}{4\pi^2}\int dk_x\, e^{ik_x(x-x')}\underbrace{\int dk_y\, \frac{e^{ik_y(y-y')}}{k_y - ik_x^2}}$$





$$\begin{cases} 2\pi i\, e^{-k_x^2(y-y')} & \\ 0 & (y<y') \end{cases}$$

$(y>y')$ $\quad G_s = -\frac{1}{2\pi}\int dk_x\, e^{ik_x(x-x') - k_x^2(y-y')} = -\frac{1}{\sqrt{4\pi(y-y')}}\, e^{-\frac{(x-x')^2}{4|y-y'|}}$

c) $\frac{\partial^2 u}{\partial x^2} - \frac{1}{k^2}\frac{\partial u}{\partial t} = 0 \quad (t>0)$

$$G_s(x,t;x',t') = -\theta(t-t')\frac{1}{\sqrt{4\pi k^2|t-t'|}}\, e^{-\frac{(x-x')^2}{4k^2|t-t'|}}$$

初期条件 $u=0\ (t<0)$, $u=f(x)\ (t=0)$, $u=?\ (t>0)$

$$u(x,t) = \int_{-\infty}^{\infty} dx'dt'\, G_s(x,t;x',t')\, f(x')\,\delta(t')$$

$$= \theta(t)\frac{1}{2k\sqrt{\pi t}}\int_{-\infty}^{\infty} dx'\, e^{-\frac{(x-x')^2}{4k^2 t}}\, f(x')$$

拡張

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{k^2}\frac{\partial u}{\partial t} = \delta(t)\, g(x)$$

$t=-\varepsilon\sim\varepsilon$ で積分 → $g(x)$ が決まる

$$\int_{-\varepsilon}^{\varepsilon} dt\, \frac{\partial^2 u}{\partial x^2} - \frac{1}{k^2}\int_{-\varepsilon}^{\varepsilon} dt\, \frac{\partial u}{\partial t} = \underbrace{\int_{-\varepsilon}^{\varepsilon} dt\, \delta(t)\, g(x)}_{1}$$

$$u(x,\varepsilon) - u(x,-\varepsilon) = 0$$

$$-\frac{1}{k^2} f(x) = g(x)$$

$t\to 0$ のふるまいを調べるには

$$\delta(x) = \lim_{\alpha\to\infty} \alpha f(\alpha x) \Big/ \int_{-\infty}^{\infty} f(x)\, dx$$

☺ $\int_{-\infty}^{\infty} \alpha f(\alpha x) g(x)\, dx = \int_{-\infty}^{\infty} f(y)\, g\left(\frac{y}{\alpha}\right) dy \xrightarrow[(\alpha\to\infty)]{} g(0)\int_{-\infty}^{\infty} f(y)\, dy$

$f(x) = e^{-x^2}$ とすると $\delta(x) = \lim_{\alpha\to\infty}\frac{1}{\sqrt{\pi}}\alpha e^{-\alpha^2 x^2}$

$\alpha = \frac{1}{2k\sqrt{\pi t}} \xrightarrow[(t\to 0)]{} \infty$ とすると $\lim_{x\to\infty}\frac{1}{2k\sqrt{\pi t}} e^{-\frac{x^2}{4k^2 t}} = \delta(x)$

$\frac{\partial^2 u}{\partial x^2} - \frac{1}{k^2}\frac{\partial u}{\partial t} = -\frac{1}{k^2} f(x)\,\delta(t)$ を Green 関数を使って解く」

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d) $G_s = \frac{1}{(2\pi)^2} \int \frac{e^{i\vec{k}\cdot(\vec{r}-\vec{r}')}}{k_x^2 + k_y^2} d\vec{k} = \frac{1}{(2\pi)^2} \int \frac{e^{ikR\cos\theta}}{k^2} k\,dk\,d\theta = \frac{1}{2\pi} \int_0^\infty \frac{dk}{k} \underbrace{\int_0^{2\pi} d\theta \frac{e^{ikR\cos\theta}}{2\pi}}_{J_0(kR)}$

$$J_0(kR) = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{kR}{2}\right)^{2n} \qquad G_{Ts} = G - \frac{1}{2\pi} \int_0^\infty \frac{dk}{k}$$

$$J_n(z) = \frac{1}{\pi i^n} \int_0^\pi e^{iz\cos\theta} \cos(n\theta)\, d\theta \quad \text{(Hansen 1/4 I')}$$

1° $G(R) = G_s(R) +$ (R independent)